

1. Overall Introduction

Raytac's MDBT50Q & MDBT50Q-P is a BT 5.2 stack (Bluetooth low energy or BLE) module designed based on **Nordic nRF52840 SoC solution**, which incorporates: **GPIO, SPI, UART, I2C, I2S, PMD, PWM, ADC, NFC** and **USB** interfaces for connecting peripherals and sensors.

Features:

1. Embedded 2.4GHz transceiver supports Bluetooth 5.2 ( **Bluetooth®**), IEEE 802.15.4 ( **THREAD** & Zigbee) & 2.4Ghz RF & ANT+ upon customer's preference.
2. Compact size with (L) 15.5 x (W) 10.5 x (H) 2.05 or 2 mm.
3. Low power requirements, ultra-low peak, average and idle mode power consumption.
4. Be compatible with a large installed base of mobile phones, tablets and computers.
5. Fully coverage of BLE software stack.
6. BLE & RF transmission switching helps products fit all operation system and most hardware.

1.1. Application

- IoT Networks
 - Smart home (such as door locks, lighting) sensors and controllers
 - Smart city sensor networks
 - Industrial IoT sensors and controllers
 - Connected white goods
- Personal Area Networks
 - Health / fitness sensor and monitor device
 - Medical device
- Interactive entertainment devices
 - Advanced remote controls
 - Gaming controller
- Advanced wearables
 - Connected watches
 - Advanced personal fitness devices
 - Wearables with wireless secure payment
 - Connected Health
 - Virtual/Augmented Reality applications
- High performance HID Controllers
- Mesh Network